

Predicting PATIENT NO-SHOWS in Healthcare

THE BRIEF

Building a machine-learning pipeline in KNIME to forecast which patients will miss their appointments — and giving hospitals a sharper tool to schedule, staff, and serve more effectively.

FIELD

Healthcare Analytics

TOOL

KNIME Analytics Platform

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Problem Statement

A 20.2% no-show rate across 110,527 appointments represents significant lost capacity in healthcare delivery.

Impact of Missed Appointments

- Waste hospital resources — staff time, equipment, and facilities go unused
- Increase waiting times for other patients seeking care
- Mean fewer patients are seen each day, and clinicians are left waiting on patients who never arrive
- Create downstream scheduling bottlenecks across departments.

Project Goal

Identify and analyze key factors that influence whether patients attend or miss scheduled medical appointments by examining demographic, temporal, and behavioral variables — and build a predictive system to flag potential no-shows in advance.

Dataset Overview

Source Hospital appointment dataset for analyzing and predicting patient no-show behavior.

Total Records 110,527 appointments

Attendance 88,208 attended (~79.8%) vs 22,319 missed (~20.2%) — No-show rate ≈ 20.2%

Features by Category

- **Patient:** Age, Gender
- **Medical:** Hypertension, Diabetes, Alcoholism, Handicap
- **Appointment:** ScheduledDay, AppointmentDay
- **Socioeconomic:** Scholarship
- **Communication:** SMS_received
- **Target Variable:** No-show (Yes / No)

Research Questions

- What factors influence whether a patient attends or misses an appointment?
- How does waiting time between scheduling and appointment affect no-show rates?
- Do certain neighbourhoods exhibit higher no-show rates than others?

- How does age influence appointment attendance?
- Do SMS reminders reduce no-show rates?

Data Preparation

- The dataset was cleaned and prepared to ensure it was suitable for analysis and modeling.
- Irrelevant columns such as PatientID and AppointmentID were removed, as they do not contribute to prediction.
- The dataset was checked for missing, inconsistent, and invalid values, and necessary corrections were applied.
- Date variables (ScheduledDay and AppointmentDay) were standardized into proper formats.

To support analysis:

- A new variable, waiting_days, was created by calculating the difference between scheduling and appointment dates.
- Patients were grouped into age categories (Children, Adults, Seniors).
- Waiting time was categorized into meaningful intervals (Same day, Within 7 days, 1–4 weeks, Over 1 month).

Finally:

- Neighbourhoods with very small sample sizes were removed to ensure reliable comparisons.
- Extreme or unrealistic values were filtered out. The final dataset consisted of over 110,000 cleaned records ready for analysis and modeling.

Methodology

The analysis was conducted using the KNIME Analytics Platform through a structured workflow pipeline.

- The process began with data cleaning, where raw data was refined and validated.
- Next, feature engineering was applied to create new variables such as waiting_days, age groups, and waiting time categories.
- Exploratory Data Analysis (EDA) was then performed using aggregation and visualization techniques to identify patterns in no-show behavior.
- The dataset was further refined through filtering and validation to improve reliability.

For modeling:

- A Decision Tree model was first developed as a baseline to establish initial performance.
- An improved model using Random Forest was then implemented to capture more complex relationships in the data.
- To address class imbalance, SMOTE was applied to the training dataset to improve the model's ability to detect no-shows.

Finally:

- Model performance was evaluated using accuracy, recall, Cohen's Kappa, and confusion matrix analysis.
- Results from both models were compared to assess improvement.

This pipeline ensured a systematic approach from raw data to actionable insights.

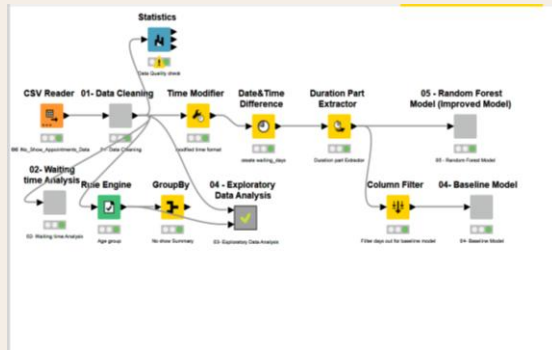


Figure X: KNIME Workflow for Data Preparation, Analysis, and Modeling.

Node colors represent processing stages in the KNIME pipeline; clarity may vary depending on export resolution.

Exploratory Data Analysis and Insight

Exploratory Data Analysis was conducted to identify patterns and relationships between patient characteristics and appointment attendance. The analysis focused on key variables including waiting time, age, neighbourhood, and SMS reminders, all of which were examined in relation to no-show behavior.

The objective of this stage was to understand how these factors influence whether patients attend or miss their scheduled appointments, and to generate insights that would inform the predictive modeling phase.

The charts below present the distribution of no-show rates across these variables, highlighting significant trends and variations within the dataset.

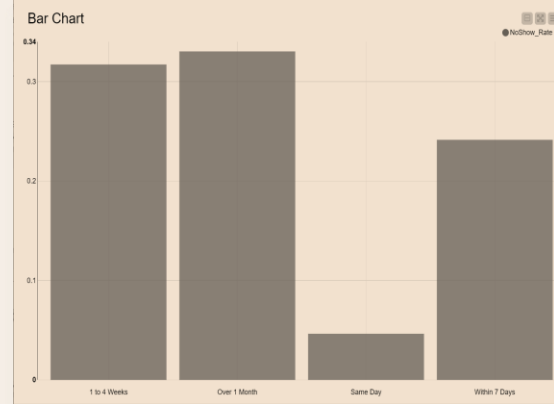


Figure 1: No-Show Rate by Waiting Time

The chart shows a clear upward trend in no-show rates as waiting time increases. Patients with same-day appointments have the lowest no-show rate, while those scheduled more than one month in advance have the highest. This indicates that longer waiting times are strongly associated with higher no-show rates, as longer delays significantly increase the risk of missed appointments. This is our strongest predictor.

- Same Day: ~5% no-show
- Within 7 Days: ~24%
- 1 to 4 Weeks: ~32%
- Over 1 Month: ~33%
- *Longer waits dramatically increase no-shows*

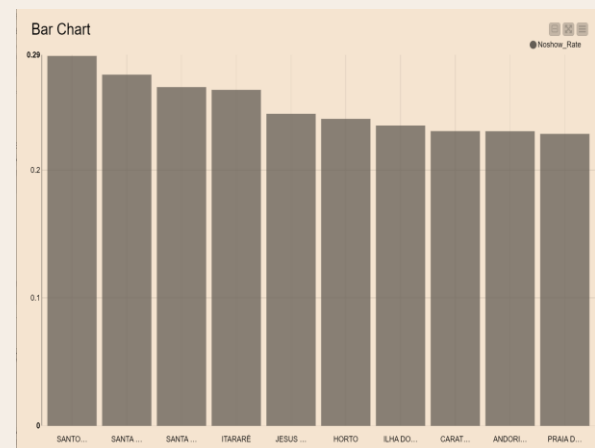


Figure 2: Top 10 Neighbourhoods by No-Show Rate

The chart shows significant variation in no-show rates across neighbourhoods. Areas such as Santos Dumont and Santa Cecília exhibit the highest rates, approaching 30%. This indicates that no-show behavior varies significantly

by location. This variation may be influenced by socioeconomic factors, access to healthcare, or transportation challenges, indicating that location-specific strategies may be necessary.

- Significant location-based variation
- High no-show: Santos Dumont 28.9%, Santa Cecília 27.5%, Itararé 26.3%
- Lower-risk: Jardim Camburi (~18.9%) despite high volume
- *Suggests location-specific interventions*

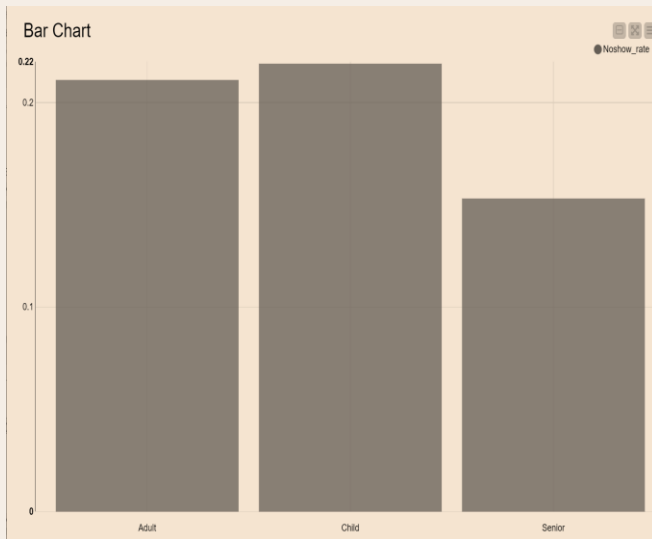


Figure 3: No-Show Rate by Age Group

The chart shows that no-show rates vary across age groups. Children exhibit the highest no-show rate, followed closely by adults, while seniors have significantly lower rates. This pattern may reflect differences in dependency and scheduling responsibility, where younger patients rely on guardians, increasing the likelihood of missed appointments.

- Children: 21.9% no-show (5,997 / 27,380)
- Adults: 21.1% no-show (13,080 / 61,974)
- Seniors: 15.3% no-show (3,242 / 21,173) — most **reliable**

Younger patients are the priority intervention group

Pred: Yes

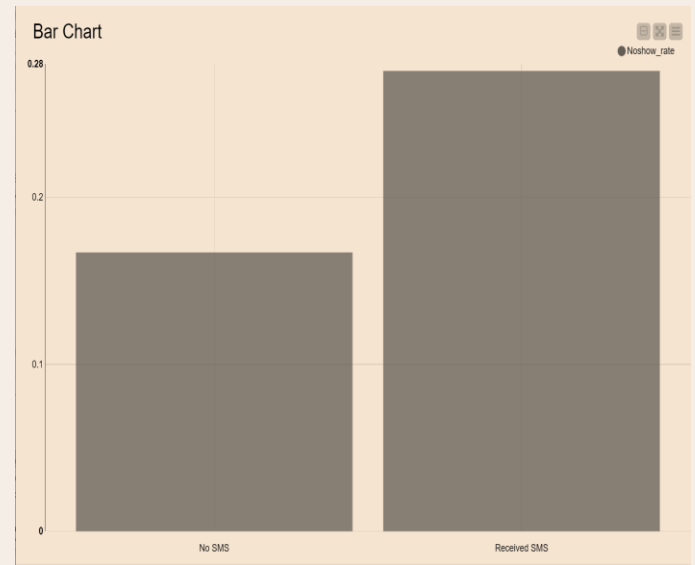


Figure 4: No-Show Rate by SMS Reminder Status

This chart compares no-show rates between patients who received SMS reminders and those who did not. Patients who received SMS reminders exhibit a higher no-show rate. This suggests that reminders are likely being used reactively for patients already identified as high-risk, rather than serving as a preventive measure, highlighting an opportunity for earlier intervention, indicating the presence of selection bias.

- No SMS: 16.7% no-show
- Received SMS: 27.6% no-show
 - Likely selection bias — SMS sent to high-risk patients
 - *SMS_received is a risk indicator, not a causal factor*

Baseline Model

Algorithm: Decision Tree (KNIME)

Top split: waiting_days ≤ 12 vs > 12

Case 1 — Waiting ≤ 12 days

95.3% show | 4.7% no-show — *highly reliable*

Case 2 — Waiting > 12 days

71.5% show | 28.5% no-show — *high risk*

CONFUSION MATRIX		
	Pred: No	Pred: Yes
Actual: No	41,404	2,700
Actual: Yes	9,641	1,519

Performance Metrics

n = 55,264 test samples

Overall Accuracy: 77.7%

Recall — Show (majority): 93.9% - Recall (No-Show = No) Strong — attendance detected

Recall — No-Show (minority): 13.6% - Recall (No-Show = Yes) Poor — misses most no-shows

Cohen's Kappa: 0.098
Weak agreement beyond chance — model barely outperforms random guessing on the no-show class

Interpretation

The model is heavily biased toward predicting attendance (the majority class). While overall accuracy appears moderate at 77.7%, it captures only 13.6% of actual no-shows — meaning 86.4% of patients who will miss their appointment go undetected. Cohen's Kappa of 0.098 confirms near-chance agreement beyond the majority-class baseline.

This baseline Decision Tree is **not suitable for deployment** without further improvement. The failure to identify no-shows — the very outcome the clinic needs to anticipate — motivates the next step: **Random Forest + SMOTE** to address class imbalance and improve minority-class recall.

Random Forest

To improve prediction performance, a Random Forest classification model was implemented using the KNIME Analytics Platform. Random Forest was selected due to its ability to reduce overfitting and capture complex relationships between variables such as waiting time, age, and neighbourhood.

To address class imbalance in the dataset (approximately79.8% attendance vs. 20.2% no-show), the SMOTE (Synthetic Minority Over-sampling Technique)

method was applied to the training data. This ensured that the model learned patterns from both classes effectively.

The Model was trained and evaluated on a dataset of **52,925 test samples**.

Model Performance

The Random Forest model achieved an overall accuracy of **71.73%**, correctly classifying **37,961 out of 52,925 patients**.

Beyond accuracy, more meaningful evaluation metrics were considered:

- **Recall (Show class): 74.6%**
→ The model correctly identifies the majority of patients who will attend their appointments.
- **Recall (No-show class): 69.5%**
→ The model successfully detects approximately **7 out of 10 patients who will miss their appointments**, representing a significant improvement over the baseline model.
- **Cohen’s Kappa: 0.435**
→ Indicates **moderate agreement beyond chance**, showing that the model performs meaningfully better than random classification

CONFUSION MATRIX		
	Pred: No	Pred: Yes
Actual: No	17,432	5,934
Actual: Yes	9,030	20,529

From this:

- **True Positives (No-shows correctly predicted): 20,529**
- **False Negatives (Missed no-shows): 9,030**
- **True Negatives (Correct attendance predictions): 17,432**
- **False Positives (Incorrect no-show predictions): 5,934**
This demonstrates that the model is able to capture a substantial proportion of no-show cases, which is critical for operational decision-making.

Key Improvements Over Baseline Model

Compared to the baseline Decision Tree model, the Random Forest model shows significant improvements:

1. Improved No-Show Detection

Recall for the no-show class increased from approximately **13.6% to 69.5%**, representing a major enhancement in identifying high-risk patients.

2. Balanced Classification

The model is no longer heavily biased toward predicting attendance, allowing for more reliable identification of both classes.

3. Stronger Predictive Power

Cohen’s Kappa improved from near-random performance to moderate agreement, confirming that the model captures meaningful patterns in the data.

4. Trade-Off in Accuracy

While overall accuracy decreased slightly, this reflects a deliberate shift toward better detection of no-shows — a more valuable outcome for hospital operations.

Interpretation

The Random Forest model demonstrates that predictive analytics can effectively identify patients at risk of missing appointments. By capturing patterns across demographic, behavioral, and scheduling variables, the model provides actionable insights that can be used to improve hospital efficiency.

While overall accuracy provides a general measure of performance, it is not the most critical metric in this context. The primary goal is to correctly identify patients who are likely to miss their appointments.

Therefore, recall for the no-show class is more important, as it reflects the model’s ability to detect patients who will not attend. The Random Forest model achieves a recall of 69.5% for no-shows, meaning it successfully identifies a majority of high-risk patients.

This makes the model practically useful, even with a moderate overall accuracy.

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Table "Tree Ensemble Column Statistic" - Rows: 9 Spec - Columns: 6 Properties Flow Variables						
Row ID	T	#splits (level 0)	T	#splits (level 1)	T	#splits (level 2)
Gender	0	4	10	32	68	128
Age	1	24	58	41	60	133
Scholarship	0	3	22	28	69	131
Hipertension	10	19	34	39	75	124
Diabetes	2	19	38	35	65	131
Alcoholism	16	26	31	33	70	128
Handicap	15	37	76	32	71	138
SMS_received	20	22	30	24	60	142
Waitingdays_...	36	46	63	36	62	145

Figure Y: Feature Importance (Random Forest Model)

Feature importance was derived from the Random Forest model using the frequency of variable splits across the trees. Variables that appear more frequently in decision splits are considered more influential in the model.

The results show that waiting_days is the most important feature, followed by age, handicap, and SMS_received. This indicates that longer waiting times and certain patient characteristics strongly influence the likelihood of missed appointments.

These findings are consistent with the exploratory data analysis, where waiting time was identified as a key driver of no-show behavior.

Key Findings

1. Waiting time is the strongest predictor of no-shows — same-day appointments have only a 5% no-show rate vs. over 33% for waits longer than one month.**2.SMS reminders correlate with higher no-show rates** (27.6% vs. 16.7%) — but this reflects selection bias (reminders sent to high-risk patients), not reminder ineffectiveness.

2. Younger patients miss appointments more — children 21.9%, adults 21.1% compared to seniors at 15.3%.

3. No-show rates vary significantly by neighbourhood — top areas exceed 25% (e.g. Santos Dumont at 28.9%). Geography matters.

4. Random Forest with SMOTE outperforms baseline Decision Tree by producing balanced classification across both show and no-show classes.

Business Implications

1. Fewer Missed Appointments

- Detect ~70% of likely no-shows in advance
- Fewer empty slots and better use of doctors and facilities
- Reduced revenue loss from unused appointment capacity.

2. Targeted Interventions

- Focus reminders, calls, and rescheduling on high-risk patients
- Lower operational cost compared to contacting all patients
- More effective communication leading to higher attendance.

3. Scheduling Efficiency

- Slight overbooking for high-risk time slots
- Prioritize reliable patients for critical appointments

- Reduced idle time and improved patient throughput.

4. Resource Allocation

- Match staff and equipment to expected attendance
- Reduce staff downtime
- Improve overall service delivery and resource utilization

5. Data-Driven Decisions

- Identify key drivers of no-shows (waiting time, SMS, age, neighbourhood)
- Support process improvements and policy decisions
- Enable more informed, evidence-based management

6. Patient Experience

- Shorter queues and faster service
- More predictable appointment scheduling
- Improved overall patient satisfaction

“The predictive model delivers actionable insights that can significantly reduce no-shows, improve hospital efficiency, and support data-driven healthcare management. Note: ~71% accuracy means the model is a decision-support tool, not a fully automated system”.

Limitations

1. Model Performance

- Although the model improved no-show detection, overall accuracy remains around **71.7%**, meaning errors still occur
- Around **28% of predictions are incorrect**, which can affect scheduling decisions.
- Cohen’s Kappa (0.435) shows only **moderate predictive strength**.
- Performance is still influenced by the original class imbalance in the dataset

2. Data & Methodology

- SMS reminders appear linked to higher no-show rates, but this reflects **targeted intervention**, not true causation.
- The dataset lacks important behavioral and contextual factors such as:
 - Patient attendance history
 - Income or transportation access
 - Personal or environmental constraints (e.g., weather)
- The analysis is **observational**, not experimental, so causal relationships cannot be confirmed.

- Results may not fully generalize to other hospitals or healthcare settings

3. Operational Considerations

- Misclassifications may lead to **incorrect scheduling decisions**, such as overbooking or missed opportunities
- The model should be used as a **decision-support tool**, not a fully automated system.
- Human oversight is required when applying predictions in real-world settings.

This model should be used as a decision-support tool, not a fully automated system

Recommendations

1. Operational Improvements

- Reduce the waiting time between booking and appointment, since longer delays increase no-shows
- Apply controlled overbooking in time slots with high predicted no-show risk
- Prioritize reliable patients for critical or time-sensitive appointments

2. Targeted Interventions

- Use the model to identify patients most likely to miss appointments
- Focus reminders, follow-up calls, and engagement efforts on these high-risk patients
- Offer flexible rescheduling options to reduce missed visits

3. Data & Model Enhancement

- Improve the dataset by including additional factors such as:
 - Patient attendance history
 - Distance to hospital
 - Socioeconomic indicators
- Continue improving the model through feature engineering and parameter tuning
- Explore more advanced models (e.g., boosting techniques) for better accuracy

4. Deployment Strategy

- Integrate the model into hospital scheduling systems for real-time support
- Use predictions as **decision support**, not full automation

- Regularly monitor model performance and update with new data

Conclusion

1. Summary of Findings

- A machine learning model was successfully developed to predict patient no-shows
- Waiting time and SMS reminders emerged as key influencing factors
- Initial baseline model performed poorly due to class imbalance, requiring improvement

2. Model Performance

- The Random Forest model achieved about **71.7% accuracy**
- It provided a more balanced prediction across both “show” and “no-show” cases
- Detection of no-show patients improved significantly compared to the baseline

3. Practical Value

- The model enables hospitals to anticipate missed appointments in advance
- Supports better scheduling, staffing, and resource allocation
- Encourages more data-driven decision-making in healthcare operations

FINAL INSIGHT

"While not perfect, the model offers meaningful predictive capability and should be used as a decision-support tool to improve healthcare efficiency."

References

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